

Complex Numbers



Arithmetic with complex numbers

- 1 Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.
 - 2 Find $(6 + i) - (2 - 5i)$.
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Complex roots of a quadratic

- 3 Solve $x^2 + 6x + 13 = 0$.
 - 4 Solve $2x^2 - 4x + 10 = 0$.
 - 5 Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.
 - 6 Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.
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Modulus

- 7 Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).
- 8 Find $|z|$ for $z = 5 - 12i$.
- 9 Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.